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A techno-economic analysis of cross-regional renewable hydrogen supply routes in China

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HIGHLIGHTS

- The techno-economics of six renewable hydrogen supply routes for hydrogen refueling stations are studied in China's context.
- The cost impacts of electrolyzer selection, transportation distance, and electricity price are explored.
- Natural gas pipeline H_2 blending route offers the lowest cost at 2000 km distance.
- Gas-hydrogen pipeline route's cost is strongly influenced by its utilization rate.
- Liquid-hydrogen truck route has the best economics when the electricity price is below 0.1 CNY/kWh.

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ABSTRACT

The cross-regional renewable hydrogen supply is significant for China to resolve the uneven distribution of renewable energy and decarbonize the transportation sector. Yet the economic comparison of various hydrogen supply routes remains obscure. This paper conducts a techno-economic analysis on six hydrogen supply routes for hydrogen refueling stations, including gas-hydrogen tube-trailer, gas-hydrogen pipeline, liquid-hydrogen truck, natural gas pipeline, MeOH truck, and NH_3 truck. Furthermore, the impacts of three critical factors are examined, including electrolyzer selection, transportation distance, and electricity price. The results indicate that, with a transport distance of 2000 km, the natural gas pipeline route offers the lowest cost, while the gas-hydrogen tube-trailer route is not economically feasible. The gas-hydrogen pipeline route shows outstanding cost competitiveness between 200 and 2000 km, while it is greatly influenced by the utilization rate. The liquid-hydrogen truck route demonstrates great potential with the electricity price decreasing. This study may provide guidance for the development of the cross-regional renewable hydrogen supply for hydrogen refueling stations in China.
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Introduction

The proposed carbon peak and carbon neutral targets have accelerated the pace of China's clean energy transition, for which the promotion of renewable energy is widely believed to be one of the most crucial drivers [1]. By the end of 2022, China's installed renewable energy capacity has reached 1200 GW, of which wind and solar power exceeding 760 GW and becoming more and more prominent [2]. Although China is abundant in renewable energy sources, there is a serious geological mismatch between its power consumption and renewable energy resources (RES). RES, especially solar and wind resources, are primarily distributed in western China, while the electricity consumption center is in the east [3]. This mismatch, combined with the intermittence of RES, gives energy storage and long-distance energy transmission an increasingly significant role in raising the share of renewable energy.

Renewable hydrogen, which is produced through RES-powered water electrolysis, is regarded as the ideal large-scale renewable energy storage medium without carbon emissions [4]. Renewable hydrogen has a wide range of applications in the fields of chemical, energy, and transportation [5]. Among all the H_2 applications, the fuel cell vehicle (FCV) has gained increasing significance as it enables synergistic emission reduction benefits in the energy and transport sectors [6–10]. Lots of researchers have studied the emission mitigation potential of renewable hydrogen as a clean fuel for the transportation sector. Siyal et al. evaluated the wind generated hydrogen production potential to provide the environmentally clean, renewable, and cheap fuel to the road transport sector of Sweden [11]. The results showed that using only 2% of total local wind-generated hydrogen, gasoline consumption and CO_2 emissions can be reduced by up to 50%. Weger et al. investigated the impacts on greenhouse gas and air pollutant emissions due to a possible hydrogen-powered road transport transition in Germany [12]. It was found that an annual emission reduction of 179 $MtCO_2eq$ could be achieved with renewable-powered electrolysis. Reuβ et, al. Carried out an assessment of spatially-resolved infrastructure to analyze the resulting costs and CO_2 emissions of different H_2 supply options in Germany [13]. To achieve zero emissions from FCV, a supply of renewable hydrogen is necessary. However, the transportation of hydrogen is difficult and expensive due to its poor safety, low volumetric density, and underdeveloped delivery infrastructure [14]. Finding the best pathway to realize a low cost of hydrogen supply is of great significance for the application of renewable hydrogen in FCV [15,16].

To this end, many researchers have done research on the techno-economics of the hydrogen supply pathway for hydrogen refueling stations (HRS). They can be classified into two groups. The first group of studies focused on the direct hydrogen supply through gas-hydrogen tube-trailer, gas-hydrogen pipeline, liquid-hydrogen truck. For example, Li et al. studied the cost of compressed gas H_2 and liquid H_2 supply for transportation applications, finding that liquid H_2 shows greater cost reduction potential than gas H_2 delivery [17]. Huo et al. developed a large-scale renewable hydrogen supply model that compared the costs of different modes of H_2 transportation, including gas-hydrogen trailers, liquid-hydrogen trucks and gas-hydrogen pipelines. The results showed that the least-cost renewable hydrogen supply pathway for a medium-distance transport scenario in China is to transport liquid hydrogen by truck [18]. Husarek et al. adopted a multi-modal energy system model incorporating electricity, natural gas as well as hydrogen transportation options for Germany in 2050. It was found that connecting major demand and supply centres via pipeline is macro-economically beneficial and the pipeline network was optimised [19]. Rong et al. conducted a techno-economic analysis of hydrogen transportation from the hydrogen plant to the terminal hydrogen refueling station, considering gas hydrogen trailers, liquid hydrogen trucks, and pipelines. The three pathways were compared and the optimal economical modes under various transportation distances and hydrogen demand were identified [20]. Borsboom-Hanson et al. compared the transport costs of liquid and gaseous hydrogen by truck, rail, barge, and pipelines, showing that the only feasible gaseous hydrogen transport option for long-distance transport is pipelines [21]. Liquid hydrogen is also found to be feasible for long distance H_2 transport using rail.

The second group of studies focused on the indirect hydrogen supply pathway using carriers such as hydrogen-natural gas mixtures, ammonia (NH_3) and methanol (MeOH). Liu et al. performed an economic evaluation of transporting renewable hydrogen-natural gas mixtures over long distances utilizing natural gas pipelines in China and analyzed the main influence items [22]. Cerniauskas et al. compared the cost of transporting hydrogen in natural gas pipelines and gas-hydrogen pipelines in Germany [23]. The result showed that natural gas pipeline reassignment can reduce hydrogen delivery costs by more than 60% compared to newly built hydrogen pipelines. Di Lullo et al. deployed a process-based model to assess the hydrogen transportation cost and emission footprints in a large-scale long-distance land-based transportation scenario [24]. The results indicated that the pure hydrogen pipelines and natural gas pipeline blending were two most preferred options for lowering costs. NH_3 and MeOH also attracted great research attention due to their outstanding physical characteristics as hydrogen carriers. For instance, Gu et al. analyzed the economic and environmental aspects of green hydrogen supply routes for hydrogen refueling stations (HRS) in China, including gaseous hydrogen tube trailers, MeOH, and liquid hydrogen [25]. Sens et al. investigated the cost of a renewable hydrogen supply from well to tank for fuel cell heavy-duty vehicles in Germany. Gaseous hydrogen, liquid hydrogen, MeOH and NH_3 are considered and compared in different scenarios [26]. Li et al. performed a well-to-pump energetic, environmental, and economic assessment of MeOH and hydrogen, implying that MeOH produced by renewable sources is the most feasible near-term pathway for the transportation sector in China [27]. Nasharuddin et al. presented a techno-economic feasibility study of using NH_3 as the H_2 carrier for FCV applications [28]. The study found that NH_3 is only economically viable as a carrier for hydrogen in FCV applications in large-scale scenarios.

However, few studies have comprehensively investigated all the above potential hydrogen supply routes for HRS in China's context. In particular, a comparative analysis of the

four carriers of gaseous hydrogen (GH_2), liquid hydrogen (LH_2), methanol (MeOH), and NH_3 under a uniform framework is still a research gap. For the four carriers, the cost effectiveness of production, transportation, decomposition, and refueling varies a lot due to different cost-effecting elements. And the identification and evaluation of key elements' impacts remain obscure at present.

In view of this, the goal of this paper is to provide a comprehensive economic analysis of GH_2, LH_2, MeOH, and NH_3 as carriers in China's cross-regional H_2 supply by considering the stages of production, transportation, decomposition, and refueling. Since most of China's HRS are located in the east while renewable energy sources are particularly abundant in the northwest, this paper selects northwest China as the hydrogen production location and Shanghai as the HRS site. The techno-economic analysis is conducted on six H_2 supply routes: (1) gas-hydrogen trailer route (GH_2-trailer); (2) gas-hydrogen pipeline (GH_2-pipeline) route; (3) liquid hydrogen truck (LH_2-truck) route; (4) natural gas pipeline hydrogen blending (NGP) route; (5) methanol as hydrogen storage carrier truck (MeOH-truck) route; and (6) liquid NH_3 as hydrogen storage carrier truck (NH_3-truck) route. Besides, the impact of key factors on the cost is examined, including electrolyzer selection, transportation distance, and electricity price. Based on the results, policy recommendations regarding China's renewable hydrogen supply are proposed.

Methods and data

System definition

The system boundary of this research includes four stages: production, transportation, decomposition, and hydrogen refueling. Fig. 1 shows the major processes of different routes. The production plants for six routes are assumed to be located in the northwest region, where solar and wind energy resources are particularly abundant. The electricity used in the production stage is assumed to be RES-generated. According to data released by the China Energy Administration, the cost of renewable electricity in northwest China is about 0.3 CNY/kWh (0.044 USD/kWh) [29]. The decomposition and refueling stages are assumed to be in Shanghai, which plays a leading role in promoting HRS in China [30]. Considering the typical distance from northwest provinces to Shanghai, the distance between the northwest and Shanghai is assumed to be 2000 km [25].

The common part of the six routes is that they all begin with water electrolysis utilizing renewable energy in northwest China and end with refueling GH_2 at 70 MPa in Shanghai. However, at each stage, they differ in details. Routes 1–4 use hydrogen as their transport carrier and differ mainly in the transportation stage, where tube trailers, pure hydrogen pipelines, liquid hydrogen trucks, and natural gas pipelines are employed, respectively. Routes 5–6 utilize renewable MeOH and NH_3 as hydrogen carriers, with trucks as the transport tool. After transportation, Routes 4–6 require decomposition processes, while Routes 1–3 do not.

Economic analysis method

In order to conduct a fair cost comparison of the six routes, it is first necessary to select a standard cost metric. The total unit cost of hydrogen (C_{H2}, in USD/kg) is used for evaluation in this study. As shown in Eq. (1), it covers all the associated costs of the four stages, including production, transportation, decomposition, and hydrogen refueling.

$$C_{H2} = C_P + C_T + C_D + C_R \tag{1}$$

Where, C_P, C_T, C_D and C_R are the unit hydrogen cost in the production, transportation, decomposition and refueling stage, respectively. For Routes 5 and 6, in which hydrogen is transported indirectly through MeOH and NH_3, the unit hydrogen cost can be expressed as:

$$C_{H2} = \left(C'_P + C'_T\right) \cdot \frac{M'}{M_{H2}} + C_D + C_R \tag{2}$$

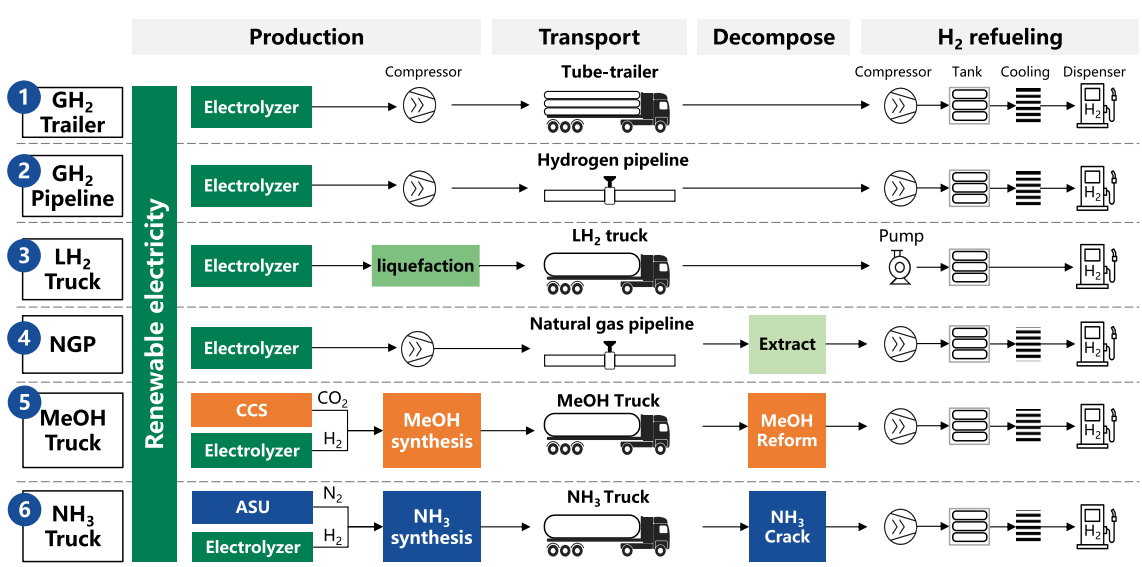

Fig. 1 – Major processes of six hydrogen supply routes.

Where, C'_P and C'_T represent the production and transportation cost of MeOH and NH_3. M' and M_{H2} mean the production rate of MeOH/NH_3 and the equivalent rate of hydrogen.

In addition to the unified cost metric, the scope of the cost analysis should be decided. In each stage, all the capital expenses and operating costs should be considered [31]. Each item in Eq. (1) can be calculated as the sum of annualized capital cost and operating cost. Taking the production cost C_P as an example:

$$C_P = C_{CAP} + C_{OP} \tag{3}$$

Where, C_{CAP} represents annualized capital cost, which can be calculated by multiplying the total capital cost (TCC) and capital recovery factor (CRF).

$$C_{CAP} = TCC \times CRF \tag{4}$$

$$CRF = \frac{i(1+i)^t}{i(1+i)^t - 1} \tag{5}$$

TCC covers all costs associated with a process unit, including the procurement of equipment, engineering permits, contingencies, working capital, and general facilities [25]. CRF indicates the present value of money over the lifetime of a process unit. It can be calculated by applying Eq. (5) and is dependent on the discount rate i and lifetime t.

As Eq. (6) shows, C_{OP} includes raw material cost, labor cost, by-product cost, and other operation and maintenance costs.

$$C_{OP} = C_{feed} + C_{labor} + C_{byproduct} + C_{O\&M} \tag{6}$$

The general economic assumptions are listed in Table 1, including the key material price, lifetime for main process units, and so on.

Technology description and related cost

Production stage

There are mainly three types of electrolyzers for water electrolysis: alkaline water electrolysis (AWE), polymer electrolyte membrane (PEM), and solid oxide electrolyzer (SOE) [34]. With a higher technology readiness level and lower capital cost, AWE is used in the majority of commercial renewable H_2 projects in China [35]. Given this background, AWE is selected as the electrolysis technology in this study. It has an efficiency of 60% and a unit power consumption of 4.8 kWh/kgH_2 [22]. The installed capacity of the H_2 production plant is assumed to be 105 MW with an annual production of 10,000 tonH_2/year. Due to the intermittency of renewable energy power, the electrolyzer can not always be in operation during its lifetime. According to the relevant literature [14,36–39], the actual operating hours of electrolyzer vary a lot with different project configuration. As Guolian securities estimated, the average operating hours in China's electrolytic hydrogen projects during 2021 and 2022 are about 4500 h per year [40]. In this paper, the electrolyzer's operating hours are assumed to be 5000 h per year considering the technology advancement. The profit from oxygen, a by-product of water electrolysis, is also considered in economic analysis. For Routes 1, 2, and 4, gaseous hydrogen only needs to be compressed during the production stage. While for Routes 3, 5, and 6, the production stage requires complex processing of gaseous hydrogen.

For Route 3, the LH_2-truck route, gaseous hydrogen must be liquefied through a hydrogen liquefaction device. Liquid hydrogen is one of the most popular hydrogen carriers, as its density of 70.8 kgH_2/m^3 is significantly greater than that of gaseous hydrogen [41]. However, the process of H_2 liquefaction is capital-intensive and power-consuming. It is achieved by cooling down the compressed hydrogen gas to 20.3 K using a cryogenic deep cooling device [41,42]. It takes approximately 16 kWh to liquefy one kgH_2 [42].

For Route 5, MeOH is used as the hydrogen carrier. The higher volumetric H_2 density (99 kg H_2/m^3) of MeOH makes it popular for large-scale long-distance delivery [43]. In this study, the MeOH is assumed to be produced through CO_2 hydrogenation. CO_2 is acquired via carbon capture and storage (CCS), and the cost of CCS is estimated to be 50 \$/tonCO_2 based on relevant literature [44–46]. The methanol synthesis from hydrogen and CO_2 consumes 0.46 kWh/MeOH [46]. Considering the practical conditions, the actual operating hours of the MeOH synthesis plant are assumed to be 7000 h per year.

For Route 6, liquid NH_3 is selected as the hydrogen carrier. In comparison to methanol, liquid NH_3 has an even higher volumetric hydrogen density of 121 kgH_2/m^3 [47]. The NH_3 is assumed to be produced by the Haber-Bosch reaction of N_2 with H_2 and then liquefied to be transported in tank trucks. The N_2 is obtained through an air separation unit (ASU), with an energy consumption of about 0.11 kWh/kg N_2 [48]. Then, the Haber-Bosch synthesis consumes about 0.53 kWh/kgNH_3 [49]. The liquefaction of NH_3 takes about 0.12 kWh/kgNH_3 [50]. Similar to MeOH synthesis, the capacity of the NH_3 synthesis plant in this study is assumed to be 1000 ton/day, and its actual operating hours are assumed to be 7000 h per year. The main cost data in the production stage of six routes are listed in Table 2.

Transportation stage

In this study, there are two modes of transportation. The first is road truck transportation, which applies to Routes 1, 3, 5, and 6. The second is pipeline transportation in Routes 2 and 4, which are H_2 pipeline transportation and natural gas pipeline, respectively.

Table 1 – General economic assumptions.

Item	Value	Unit	Refs
Electricity price	0.044	USD/kWh	[29]
Water price	0.22	USD/ton	[25]
Discount rate	5%	1	[17]
Exchange rate of Chinese Yuan (CNY)	6.80	CNY/USD	[18]
Oxygen profit	0.037	USD/Nm^3O_2	[25]
H_2 production capacity	10,000	Ton/year	/
Lifetime for H_2 production	20	Years	[27]
Lifetime for MeOH production	20	Years	[27]
Lifetime for NH_3 production	20	Years	[32]
Lifetime for H_2 pipeline	20	Years	[32]
Lifetime for H_2/MeOH/NH_3 truck	10	Years	[25,32]
HRS capacity	1000	Kg/day	[33]
Maintenance cost	2%	Of initial CAPEX	[25]
Engineering permitting etc	5%	Of initial CAPEX	[25]
Contingencies	5%	Of initial CAPEX	[25]
Working capital	3%	Of initial CAPEX	[25]
General facilities	10%	Of initial CAPEX	[25]
Transportation distance	2000	Km	[25]

Table 2 – Main cost data in the production stage.

Item	Value	Unit/spec	Refs
AWE Electrolyzer	367.6	USD/kW	[51]
AWE Stack replacement	50	% of initial CAPEX	[50]
Water-electrolysis	11	kg/kgH_2	[25]
Electricity-electrolysis	4.8	kWh/kgH_2	[22]
Electricity-compression	1	kWh/kgH_2	[22]
H_2 liquefaction plant	4.9	Million USD/(tonH_2/day)	[52]
Electricity-liquefaction	16	kWh/kgH_2	[42]
MeOH reactor	213	Million USD/(1190 tonMeOH/day)	[46]
CO_2 capture cost	50	USD/tonCO_2	[44]
Electricity-MeOH synthesis	0.46	kWh/kgMeOH	[46]
Water-MeOH synthesis	10.8	kg/kgMeOH	[27]
NH_3 reactor	$3.4 \times 10^6 \times$ (ton NH_3/day)$^{0.5}$	USD	[48]
Air separation unit	$9.2 \times 10^5 \times$ (ton N_2/day)$^{0.49}$	USD	[48]
Electricity-NH_3 synthesis	0.53	kWh/kgNH_3	[49]
Electricity-ASU	0.11	kWh/kgN_2	[49]
Electricity-NH_3 liquefaction	0.12	kWh/kgNH_3	[53]

Table 3 – Key assumptions for truck transportation in Routes 1, 3, 5, and 6.

Item	Route 1	Route 3	Route 5	Route 6
Capacity per truck	350 kgH_2	4000 kgH_2	20 tonMeOH	30 tonNH_3
Fuel consumption (L/100 km)	25	29	29	29
Fuel type	Diesel			
Fuel price	7 CNY/L (1.03 USD/L)			
Driver cost	8000 CNY/Month (1176.5 USD/Month)			
Road toll	0.5 CNY/km (0.07 USD/km)			
Average speed	70 km/h			
Operating hours	16 h/day			
distance (km)	2000			

For road truck transportation in Routes 1, 3, 5, and 6, the capacity, fuel consumption, and speed of the trucks are listed in Table 3. It is worth noting that Route 1 uses 20 MPa gaseous hydrogen tube trailers, while other routes employ tank trucks with substantially higher capacities. Other than transport capacity and fuel consumption, the remaining parameters in Table 3, such as fuel type, are assumed to be the same for all four routes. Specifically, the loss of vaporization and dissipation of liquid hydrogen must be accounted for, as its vaporization is far easier than that of other carriers. The loss rate per transportation is assumed to be 5% [54].

Hydrogen pipelines can achieve point-to-point transportation through specially constructed pipelines. In this way, gaseous hydrogen can be compressed at the production site and then transported directly to the consumption site. With comparatively low energy consumption, the pure H_2 pipeline is believed to be promising for the large-scale and long-distance transportation of hydrogen [55]. However, due to the complexity of pipeline construction and high capital costs, there are very few hydrogen pipelines (only 4600 km worldwide until the end of 2022 [56]). In this study, it is assumed that hydrogen pipelines are newly constructed and connect the hydrogen production site to the consuming location. Thus, the transportation cost per kg of H_2 depends mainly on the pipeline construction cost and the H_2 volume actually transported. Generally speaking, the larger the transportation volume, the lower the unit cost of hydrogen transportation. To calculate a 2000-km-long pipeline's construction cost, we use the "Jiyuan-Luoyang" hydrogen pipeline in China as a benchmark [32]. The "Jiyuan-Luoyang" hydrogen pipeline has a diameter of 500 mm and a service life of 20 years. The length of the pipeline is 25 km and its total capital investment is about 21.5 million USD [57]. It can transport 100,000 tons of H_2 per year with a 4.6% loss rate [32]. In addition to the initial pipeline construction CAPEX, the maintenance costs of the hydrogen pipeline are also considered, which account for about 8% of the initial CAPEX [57].

In the NGP route, gaseous hydrogen is blended into the existing natural gas networks and then transported as hydrogen-natural gas mixtures. After transportation, the hydrogen is extracted from hydrogen-natural gas mixtures. By leveraging the existing natural gas pipeline network in China, the high CAPEX of hydrogen pipeline construction can be avoided. Our economic analysis is based on Liu's study of large-scale hydrogen NGP transportation in China with a blending ratio of 5.8%, at which the NGP does not require retrofitting [22]. The Ningxia-Shanghai NGP is selected in our scenario, with a transportation toll of 0.03 USD/($Nm^3 \cdot$1,000 km) reported by China National Petroleum Eastern Pipeline Corporation [58]. The cost data of main transport equipment in six routes are shown in Table 4.

Decomposition stage

After the transportation stage, Routes 1–3 (GH_2-trailer, GH_2-pipeline, and LH_2-truck) can deliver pure hydrogen without further decomposition and purification. For other routes,

Table 4 – Cost of main equipment in the transportation stage.

Item		Value	Unit	Refs
Route 1	Tube-trailer	0.18	Million USD	[59]
Route 2	Pipeline construction	0.86	Million USD/km	[59]
	Pipeline maintenance	8%	Of initial cost	[32]
Route 3	Liquid H_2 truck cab	0.15	Million USD	[25]
	Liquid H_2 tank trailer	0.92	Million USD	[25]
Route 4	NGP-transport toll	0.03	USD/(Nm^3·1000 km)	[58]
	Compressor	296.3	USD/(Nm^3/h)	[22]
Route 5, 6	NH_3/MeOH tank truck	0.15	Million USD	[60]

decomposition is required to obtain pure hydrogen. The purity of hydrogen obtained from all routes is greater than 99.97%, which meets the requirements of China's relevant standards [61].

The Route 4 (NGP) uses pressure swing adsorption (PSA) to separate hydrogen from hydrogen-natural gas mixtures. PSA is a widely-used technology for purifying hydrogen and can achieve high purity hydrogen under a variety of operating situations [62]. The capital and operation costs of PSA are calculated based on Liu's study [22].

In Route 5, MeOH needs to be reformed to hydrogen through Methanol Steam Reforming (MSR). MSR is the reaction of MeOH with water to produce carbon dioxide and hydrogen [63]. The main devices of MSR include the membrane module, compressor [63]. After MSR, the purification of hydrogen through PSA is necessary in order to achieve the high purity of hydrogen required for FCV. In Route 6, the NH_3 can be cracked to produce hydrogen. Also, the separation and purification of H_2 are necessary for the use of FCV after the cracking. We adopt the cracking system developed by Li, which uses the PSA-to-membrane system to produce high-purity hydrogen [64]. Given that the actual decomposition process of MeOH and NH_3 results in some hydrogen loss, the recovery rate for Routes 5 and 6 is set at 95% [63,64].

The specific parameters for decomposition cost are listed in Table 5. It is worth noting that the values have been further processed based on the relevant study for a uniform and concise presentation.

H_2 refueling stage

Considering the promotion plan of HRS in China, the refueling capacity of HRS in Shanghai is assumed as 1000 kg/day with a dispensing pressure of 70 MPa [33,65]. Although the state of hydrogen refueled into FCV is the same in all six routes, the hydrogen handling process at HRS varies due to the different hydrogen supply forms before refueling. For gaseous hydrogen supply, the refueling stage contains multi-stage pressurization, cascade storage, pre-cooling, and dispensing. Different hydrogen supply pressures have an influence on the pressurization. Except for Route 1, where the hydrogen transport pressure of the trailer is 20 MPa, the pressure of the gas-hydrogen before refueling in other routes is 2 MPa, meaning a more demanding pressurization and a higher cost compared to that of 20 MPa.

Table 5 – Key cost data of decomposition.

Route	Item	Value	Refs
Route 4	H_2 extraction-PSA	4.49 CNY/kgH_2 (0.66 USD/kgH_2)	[22]
Route 5	MeOH reformer	14,813 USD/year (700 m^3H_2/h)	[63]
	Materials and utilities	0.88 USD/kgH_2	[63]
Route 6	NH_3-Craker	0.72 USD/kgH_2	[64]
	Materials and utilities	0.53 USD/kgH_2	[64]

Table 6 – H_2 refueling costs for different hydrogen supply.

Item (USD/kgH_2)	Liquid H_2	Gaseous H_2 (2 MPa)	Gaseous H_2 (20 MPa)
CAPEX	1.31	1.94	1.79
OPEX	0.95	0.94	0.90
Total Cost	2.26	2.88	2.69

The liquid hydrogen refueling station has a different configuration from the gas hydrogen refueling station. For the LH_2-truck route, hydrogen refueling includes processes of cryogenic pumping, evaporation, and dispensing. Since hydrogen gasification is a strong heat-absorbing reaction, liquid hydrogen refueling stations do not need the pre-cooling device. The avoidance of pre-cooling devices and multi-stage compressors can help reduce the refueling cost [17].

The cost of the H_2 refueling in this study is calculated based on the Argonne National Laboratory's HRSAM model [66]. By setting different HRS parameters, the unit H_2 refueling cost is calculated as shown in Table 6.

Results and discussion

Techno-economic analysis results

By summing up the unit H_2 cost of four stages, the economic analysis results of six routes are shown in Fig. 2. With an electricity price of 0.3 CNY/kWh (0.044 USD/kWh) and a transportation distance of 2000 km, the total H_2 cost of each route is ranked as follows: GH_2-trailer, MeOH-truck, NH_3-truck, LH_2-truck, GH_2-pipeline, and NGP. The total H_2 cost of Route 1, GH_2-trailer, is significantly higher than others due to the huge transportation cost. The NGP route offers the lowest total H_2 cost, whereas the GH_2-pipeline route is close to it. The three routes of LH_2-truck, NH_3-truck, and MeOH-truck have comparable total costs, with 8.09, 8.20, and 8.24 USD/kgH_2, respectively.

The pie chart in Fig. 2 presents the proportion of six routes' most costly stage (indicated by color). Except for the GH_2-

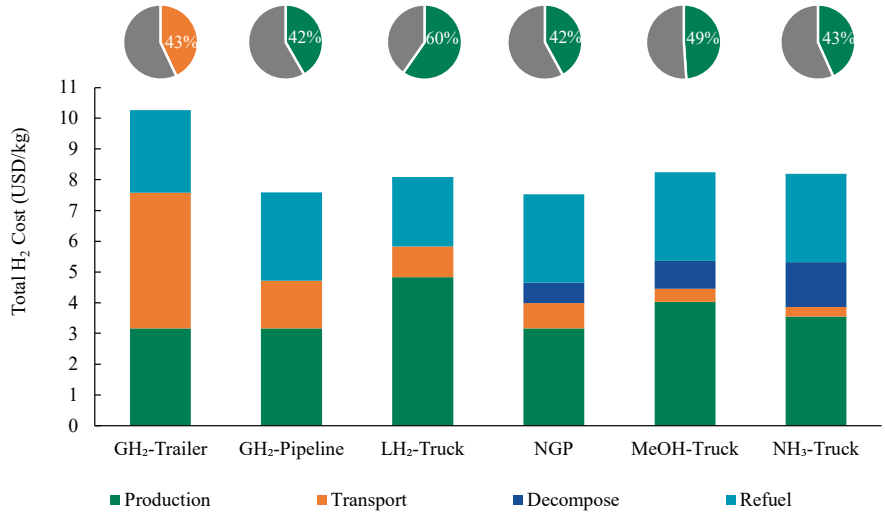

Fig. 2 – Total H_2 cost of six routes and the proportion of each route's most important stage.

trailer route, of which transportation costs account for the largest share (43%), the most expensive of the other five routes is the production stage. Particularly, the production of LH_2 is the most expensive among the six routes, accounting for sixty percent of LH_2-truck's total H_2 cost. All six routes have a share of the production stage that is higher than 40%, demonstrating that the production stage plays a crucial role in the life cycle regardless of the route. For MeOH and NH_3, other than water electrolysis, the synthesis cost in the production stage is also considerable. Due to the CO_2 capture cost, MeOH has a higher synthesis cost of 0.86 USD/kgH_2 than 0.39 USD/kgH_2 for NH_3.

Regarding the transportation stage, the transportation cost of each route varies greatly. The NH_3-Truck route offers the lowest transportation costs at 0.31 USD/kgH_2, followed by the MeOH-truck route at 0.43 USD/kgH_2. This is mainly because trucks transporting NH_3 have a larger capacity than trucks transporting MeOH, resulting in a lower unit transportation cost for NH_3. The transportation costs account for only 5% and 4% of total H_2 costs in the MeOH-truck and NH_3-truck routes, showing their economic advantage in long-distance hydrogen delivery. In contrast, the GH_2-trailer route has an incredibly high transportation cost of 4.42 USD/kgH_2, which is even higher than the H_2 production cost. The cost of H_2 refueling is significant for all routes, with the lowest cost for liquid hydrogen supply at 2.26 USD/kgH_2 and 2.88 USD/kgH_2 for gas hydrogen supply.

To validate our research, we compared the results with similar existing studies. Fig. 3 shows the comparison of the total H_2 supply cost for different studies. It is worth noting that the cost of hydrogen supply varies a lot across different scenarios and study scopes. Factors, such as the electricity price, transportation distance, and plant configuration, can have an impact on the cost of hydrogen. However, the results of this paper lie within the reasonable range of previous research results. As indicated in Fig. 3, the results of the GH_2-Trailer and NGP routes differed significantly from previous studies. For the GH_2-Trailer route, the cost of this study is significantly

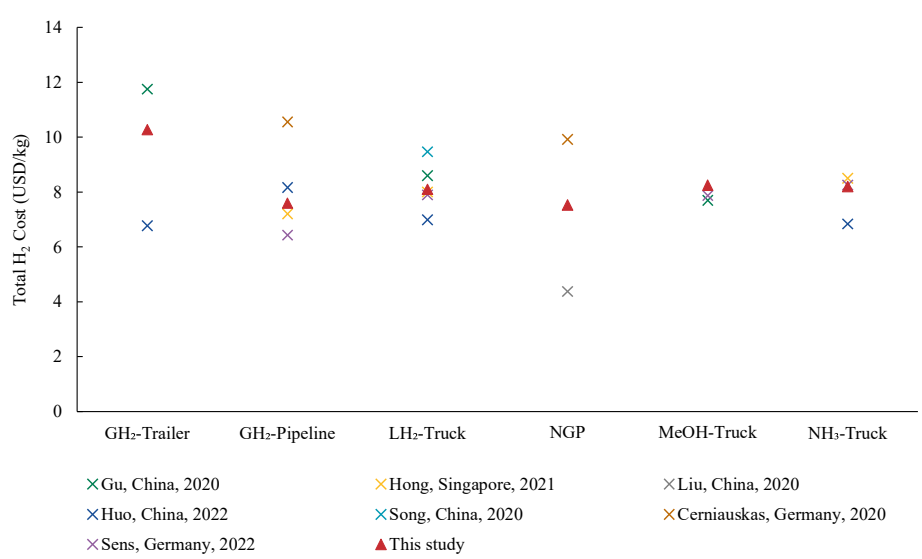

Fig. 3 – Comparison of the total H_2 cost with existing studies [22,25,26,53,54,67,68].

Fig. 4 – Cost structure of six H_2 supply routes.

higher than that of Huo's study, mainly because Huo's study focuses on the short-distance transportation scenario (300 km) [54]. For the NGP route, the cost difference mainly comes from the different study scope. Liu's study did not include the refueling stage, while Cerniauskas considered the cost related to natural gas pipeline reassignment [22,23].

Further, the cost structure of six routes in this study is analyzed and shown in Fig. 4. In general, OPEX dominates total H_2 cost, as its share ranges from 49% to 72% in different routes. The GH_2-trailer route has the largest OPEX share of 72% among the six routes, due to the low capacity of the tube trailer and consequently high fuel cost per kgH_2 for long-distance transportation. The GH_2-pipeline route has the lowest OPEX percentage among six routes, at 49%. The primary reason is the massive CAPEX associated with H_2 pipeline construction.

The cost structure also varies considerably between different stages. The production stage of each route is dominated by OPEX, accounting for 74–83% among the six routes. The high electricity cost for water electrolysis is the primary contributor. Since the power consumption of hydrogen production depends heavily on the electrolyzer type, the cost of electricity and the selection of the electrolyzer are two significant factors affecting the production cost. In contrast, CAPEX is more important in the refueling stage with a share of 58–67% among six routes. This is caused by the high equipment capital cost, which is also consistent with the findings of previous studies [17,25,69].

Effects of key factors on total H_2 cost

The above comparison among the strategies is based on the baseline assumptions. Based on the cost analysis of the six routes, electricity cost, electrolyzer choice, transportation distance, and carbon capture costs are found to be four key cost-affecting factors. In this section, their effects on H_2 costs are evaluated in depth.

Electrolyzer selection

In section Methods and data, AWE is selected for water electrolysis. As the results show, hydrogen production plays a crucial role in the total H_2 cost, and the electrolyzer selection can significantly affect the H_2 production cost. Although AWE is currently the most commonly used eletrolyzer in commercial projects, PEM and SOEC are receiving increasing attention as their technology advances. The technical specifications of AWE, PEM, and SOEC are displayed in Table 7. It can be seen that the installed cost of AWE is the lowest, while AWE has obvious limitations in efficiency and cold start time.

In this section, we select Route 4, which is the best economic route in our previous analysis, to evaluate the impact of electrolyzer selection. Assuming that the three electrolyzers have the same operating hours of 5000 per year, the total H_2 cost is illustrated in Fig. 5. The H_2 production cost using AWE, PEM, and SOEC is 3.16, 3.96, and 4.77 USD/kgH_2, accounting for

Table 7 – Specifications of three water electrolyzers [51].

Item	AWE	PEM	SOEC
Electrolyzer cost (CNY/kW)	2500–4000	10,000–15,000	19,306
Energy efficiency	60–75%	70–90%	85–94%
Power consumption (kWh/Nm^3 H_2)	4.5–5.5	3.8–5.0	3.1–3.8
Stack replacement cost	50%	25%	23%
Cold start time	Minutes	Seconds	–
Technology readiness	Commercial	Pilot	Lab

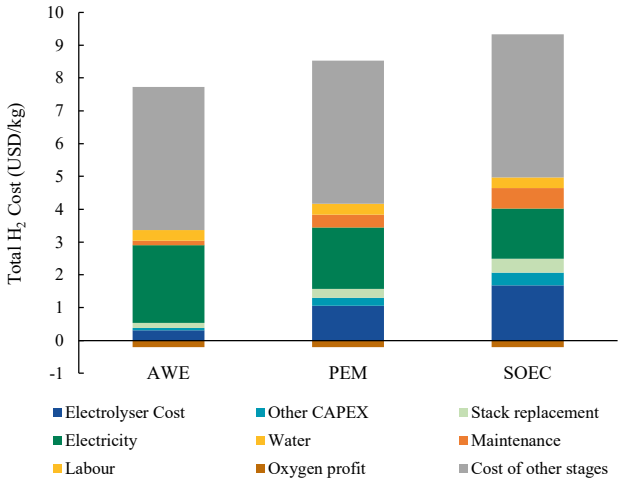

Fig. 5 – Total H_2 cost of Route 4 with different electrolyzers.

42%, 48%, and 52% of the total H_2 cost, respectively. In addition, due to AWE's lower energy efficiency, it has a significantly greater electricity cost compared with PEM and SOEC. Regarding the production stage, electricity cost accounts for 75% of AWE's hydrogen production costs, compared to 60% for PEM and 32% for SOEC.

Furthermore, the cost impact of the flexibility of three electrolyzers is examined, which corresponds to the cold start time performance. Ideally, an electrolyzer should operate for as many hours as possible for the lowest feasible cost. However, solar and wind power generation are highly variable, placing greater demands on the flexibility of RES-powered electrolyzers. As a result of its worse cold-start performance, AWE has fewer operating hours under the same RES power input compared to PEM and SOEC. To further investigate the cost impact of such AWE's disadvantages, we calculate the H_2 production cost of AWE at varying operating hours per year. Then we compare it to that of PEM and SOEC at 5000 operating hours per year, which is the baseline case in Section Methods and data.

As Fig. 6 illustrates, the red line presents the total production cost of AWE at different operating hours, which equals the sum of all the cost items in the bar chart minus the oxygen profit. The total production cost of AWE decreases significantly with the increased operating hours. This is attributable to the reduced unit CAPEX resulting from the economies of scale. However, as the operating hours increase, this benefit gradually diminishes. The conditions under which AWE approaches cost parity with PEM and SOEC are determined. When the annual operating hours of AWE are 2,300, about half those of PEM, PEM can offer a H_2 production cost roughly equivalent to that of AWE. As for cost parity with SOEC, the AWE's annual operating hours should decrease to about 30% of SOEC's.

Transportation distance

For the transportation stage, distance is one of the most important cost-affecting factors. In order to further assess the economic impact of distance, the total H_2 costs of six routes with different transportation distances are calculated.

The sensitivity of the total H_2 cost to transport distance is quite different between routes. As Fig. 7 shows, the GH_2-trailer route is the most sensitive to the transport distance. Its total H_2 cost increases by nearly 4.5 USD/kgH_2 with the distance rising from 0 to 2000 km. Specially for the GH_2-pipeline route, the transportation cost is greatly affected by the utilization rate, as the actual transport volume is a major determinant of unit capital cost for pipeline construction. Thus, in addition to the six routes defined in Section Methods and data, we further examine the impact of utilization rate and establish a low-utilization scenario for the GH_2-pipeline route (GH_2-pipeline-

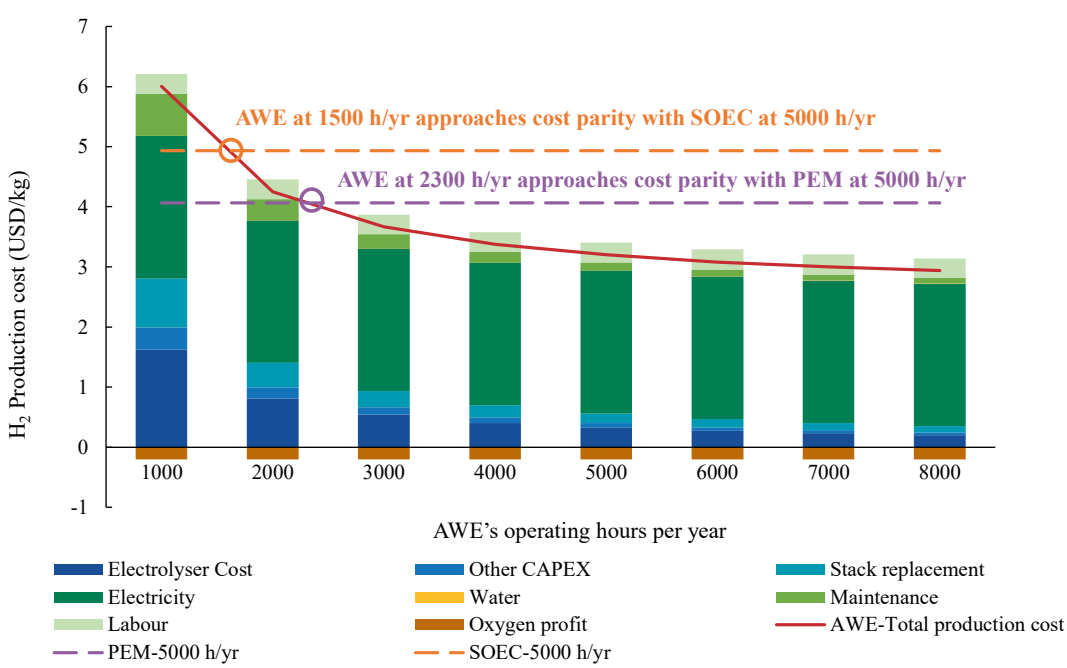

Fig. 6 – H_2 production cost of AWE at varying operating hours, compared with that of PEM and SOEC at 5000 h/yr.

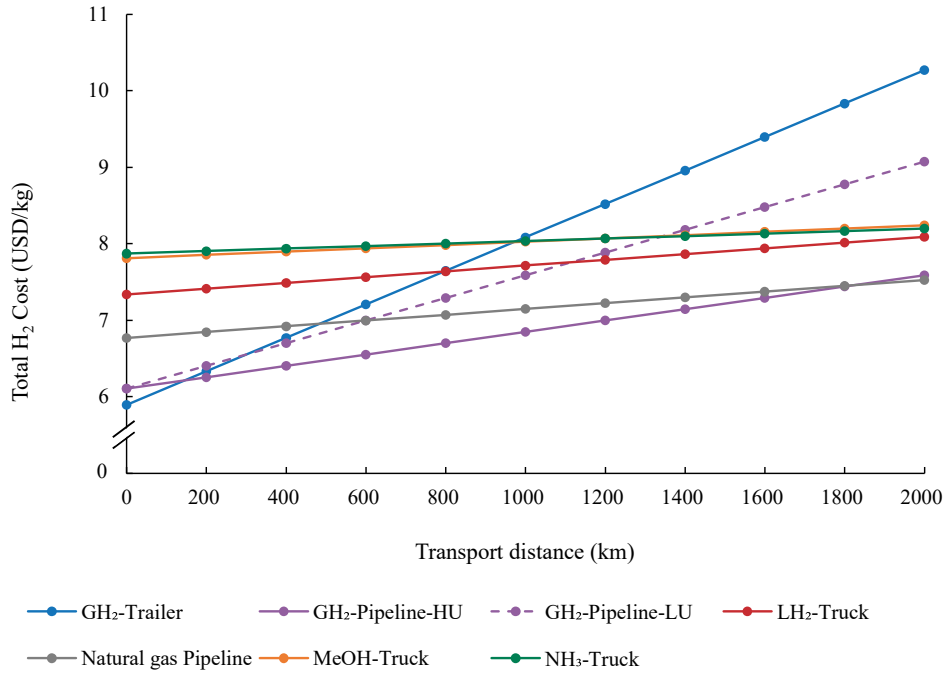

Fig. 7 – Effect of transport distance on total H_2 cost.

LU), with an actual H_2 transport volume equal to 50% of its capacity. The GH_2-pipeline-HU stands for pipeline transport with a high utilization rate (100% of its capacity). It can be seen that the cost gap between two utilization scenarios is dramatic, and the GH_2-pipeline with the lower utilization rate demonstrates a significantly high sensitivity to transport distance. For truck transport of LH_2, NH_3 and MeOH, the total H_2 cost stays comparatively stable with the varying distance. Among the six routes, the NH_3-truck route is the least sensitive to transport distance, followed by the MeOH-truck route.

For short-distance transportation (less than 200 km), the GH_2-trailer is the most economical route, but once the distance exceeds 1000 km, it becomes the most expensive option, which is consistent with the findings of previous research [38]. For distances between 200 and 1800 km, the GH_2-pipeline-HU route offers the lowest total H_2 cost. However, such a low cost needs a huge transportation volume to dilute the enormous pipeline construction CAPEX. While long-distance and large-volume point-to-point hydrogen transportation has high requirements for site selection, planning, and end-point distribution. Once the utilization rate of the pipeline decreases, the total H_2 cost of the GH_2-pipeline route would rise steeply (as shown in the GH_2-pipeline-LU scenario). The NGP route offers the second lowest total H_2 cost for most medium and long distances after the GH_2-pipeline-HU. Similar to the GH_2-pipeline route, there are also some practical limitations to the NGP route. China has only a limited number of existing natural gas pipelines, and blending high volumes of hydrogen into NGP will inevitably sacrifice natural gas transportation volume. It is worth noting that Fig. 7 compares all the routes in terms of total H_2 cost, meaning the cost includes production, transportation, decomposition and refueling. If considering the transportation stage only, truck transportation of MeOH, NH_3, and LH_2 has lower transportation costs compared with pipelines for long-distance transportation. Truck transportation is also more flexible than pipelines, indicating its strong competitiveness in practical large-scale hydrogen transportation.

Electricity price

Fig. 8 presents the total hydrogen cost of each route at different electricity prices, keeping other parameters the same as those in Section Methods and data. The electricity price range is selected as 0–0.1 USD/kWh, which can cover the majority of renewable energy projects in China.

Among the six routes, the total H_2 cost of the LH_2-truck route is most sensitive to the price of electricity. This is attributable to the massive power consumption for hydrogen liquefaction. As the cost of renewable energy continues to decrease in the future, the economic advantage of LH_2-truck will become increasingly obvious. The cross-over points of the LH_2-truck route with the other five routes are circled in Fig. 8 (GH_2-pipleine's not shown), and the corresponding electricity prices are listed in the table. With the break-even electricity price, we can derive the cost advantages of other routes compared with liquid hydrogen under different electricity price ranges. When the electricity price is less than 0.09 CNY/kWh (0.014 USD/kWh), the total hydrogen cost of the LH_2-truck is the lowest among the six routes. For routes 5 and 6, which have similar properties to liquid hydrogen, the break-even electricity price with LH_2-truck is 0.053 and 0.056 USD/kWh, respectively. This means that when the electricity price is higher than 0.053 or 0.056 USD/kWh, NH_3 or MeOH will be more cost-effective as H_2 carriers compared to LH_2. It is worth noting that although the electricity consumption of the GH_2-trailer route is low, its total cost is much higher than the other five routes in an electricity price range of 0–0.1 USD/kWh, implying that GH_2-trailer is not economically feasible for large-scale, long-distance hydrogen transportation.

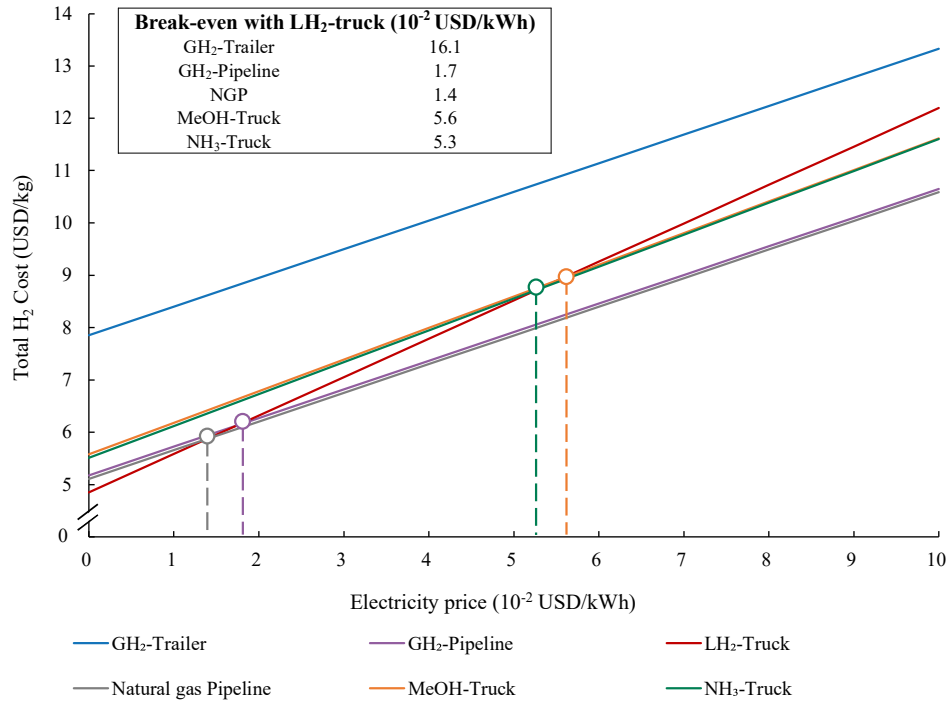

Fig. 8 – Effect of electricity price on total H_2 cost.

Policy recommendation

Based on the economic analysis of the six routes and the exploration of key cost-effecting factors, this paper puts forward the following policy recommendations for China's renewable hydrogen supply pathway from the northwest to the eastern hydrogen refueling stations.

- For renewable hydrogen production, China should put more effort into PEM's development. For example, the government could introduce an incentive policy for PEM's projects to facilitate their commercialization, especially in the northwest region. Due to its better flexibility and compatibility with renewable power generation, the PEM electrolyzer shows great potential to achieve cost parity with the AWE as renewable energy penetration continuously increases.
- The natural gas networks connecting the northwest and east should be utilized as much as possible to transport hydrogen, and the investment in a long-distance pure hydrogen pipeline should be carefully considered. For long-distance H_2 transportation and HRS applications, NGP is the most cost-effective among six routes. China should prioritize transporting hydrogen via NGP over newly built long-distance pure hydrogen pipelines. The construction cost of a long-distance H_2 pipeline is massive, and the associated risk is significant. Because the unit H_2 transportation cost will increase dramatically if the utilization rate of the newly built pure hydrogen pipeline decreases.
- In addition to pipeline transportation, carriers with high hydrogen storage density, such as liquid hydrogen, NH_3, and MeOH, should be promoted to enhance the flexibility and diversity of hydrogen transportation. Focusing on the hydrogen application of FCV, LH_2 should be prioritized over NH_3 and MeOH for the reasons listed below.
  - The LH_2-truck route has the highest cost sensitivity to the electricity price, which means it can benefit most from cost reduction of renewable energy. LH_2-truck will be the most cost-effective route among the six routes if the electricity price decreases to below 0.1 CNY/kWh (0.014 USD/kWh).
  - The LH_2-truck route offers a comparable transportation cost to MeOH-truck and NH_3-truck routes, and LH_2 can be transformed to GH_2 without complex and costly decomposition. The low-cost benefit of MeOH and NH_3 in long-distance truck transportation is counterbalanced by the high cost of cracking and purification. For example, the cost of NH_3 decomposition and purification is about 1.5 USD/kgH_2, nearly 5 times its transportation cost.
  - In the refueling stage, the different configuration of LH_2 HRS can reduce equipment requirements, allowing for considerable cost savings.

Conclusion

In this study, six routes for transporting renewable-energy based hydrogen from western China to HRS in the east have been evaluated, including GH_2-trailer, GH_2-pipeline, LH_2-truck, natural gas pipeline, MeOH-truck, and NH_3-truck. A life cycle cost analysis of six routes has been presented, which covers production, transportation, decomposition, and refueling. Based on the above analysis, the conclusions are drawn as follows.

(1) Assuming a transport distance of 2000 km and an electricity price of 0.3 CNY/kWh (0.044 USD/kWh), the total H_2 costs for the six routes are ranked as follows: GH_2-trailer, MeOH-truck, NH_3-truck, LH_2-truck, GH_2-pipeline, and NGP. For the GH_2-trailer route, transportation costs are much higher than for other routes and dominate its total H_2 cost, whereas for the other five routes, the production stage accounts for the largest share of costs.
(2) Hydrogen production cost is mainly dominated by the electricity price regardless of the electrolyzer type. When the operating hours of AWE are about 50% that of PEM, the hydrogen production cost of AWE is equivalent to that of PEM. In terms of transport distance, the optimal route is the GH_2-trailer for short distances (<200 km). At medium and long transport distances (200–2000 km), the GH_2-pipeline and NGP routes are the two most cost-effective options, with comparable total H_2 cost. Yet, both have high requirements for supply and consumption site selection. The truck transport routes of LH_2, MeOH, and NH_3 all show a relatively low cost sensitivity to the distance.
(3) The LH_2-truck route demonstrates the highest sensitivity to the electricity price and offers the lowest total hydrogen cost of the six routes when the electricity price is less than 0.014 USD/kWh. NH_3-truck and MeOH-truck have cost advantages over LH_2-truck when electricity price exceeds 0.053 and 0.056 USD/kWh, respectively.

Our results provide important insights into the economics of long-distance renewable hydrogen supply routes for HRS and the key factors' effects on the total H_2 cost in China. Nevertheless, this research has a number of limitations. First, this study does not consider the future cost reduction of related technology, especially renewable electricity. Second, for the pure hydrogen pipeline route, this paper adopts a linear fitting for cost calculation. Since there are still no existing cases of 2000-km-long hydrogen pipelines for reference, the linear cost estimation may cause some inaccuracies. Third, this paper does not consider the specifications of the FCV market at destination and focuses only on the hydrogen supply pathway, which may reduce the applicability of the results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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